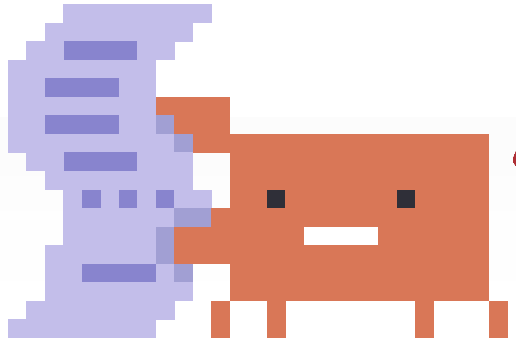
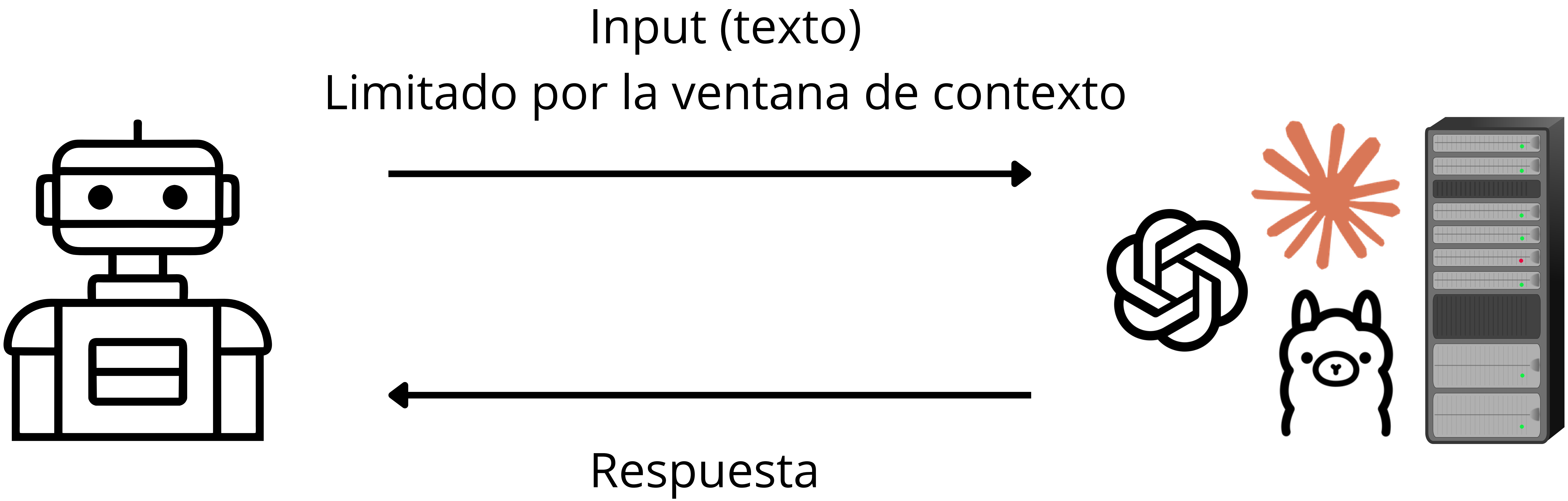
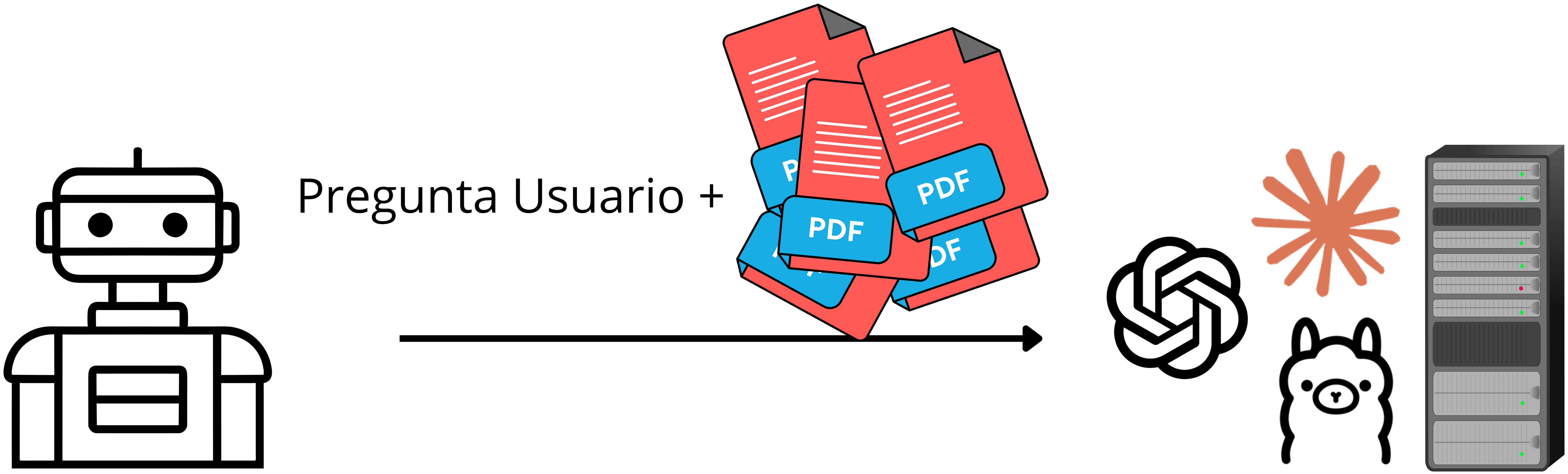
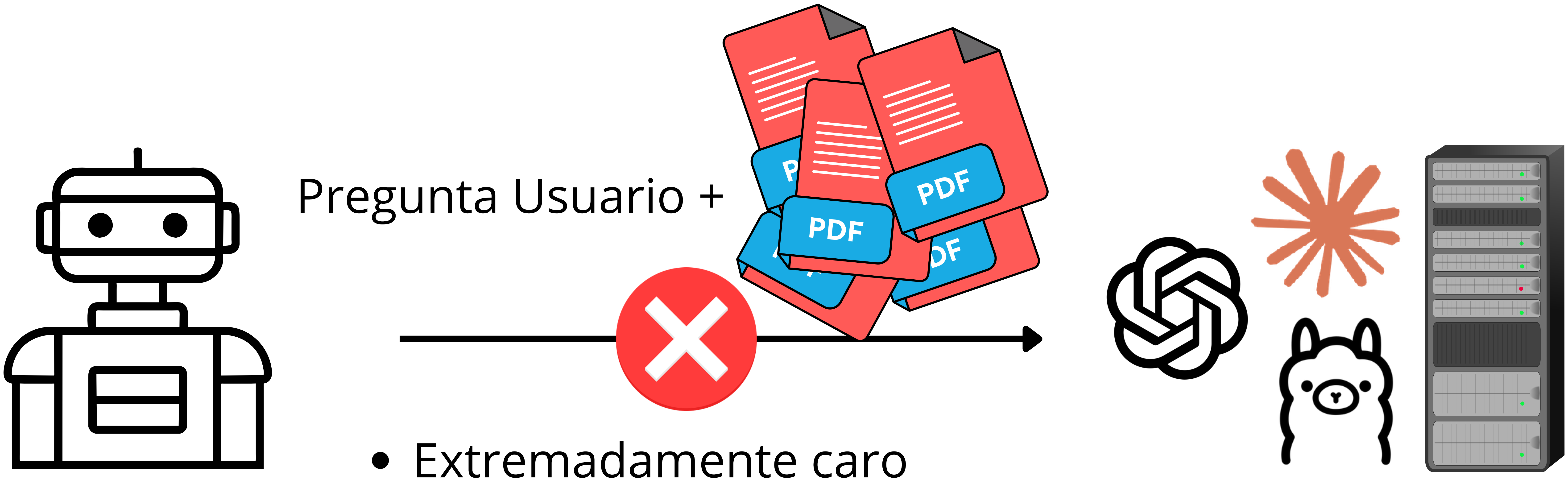
// conectar modelos a contexto y herramientas

RAG + MCP

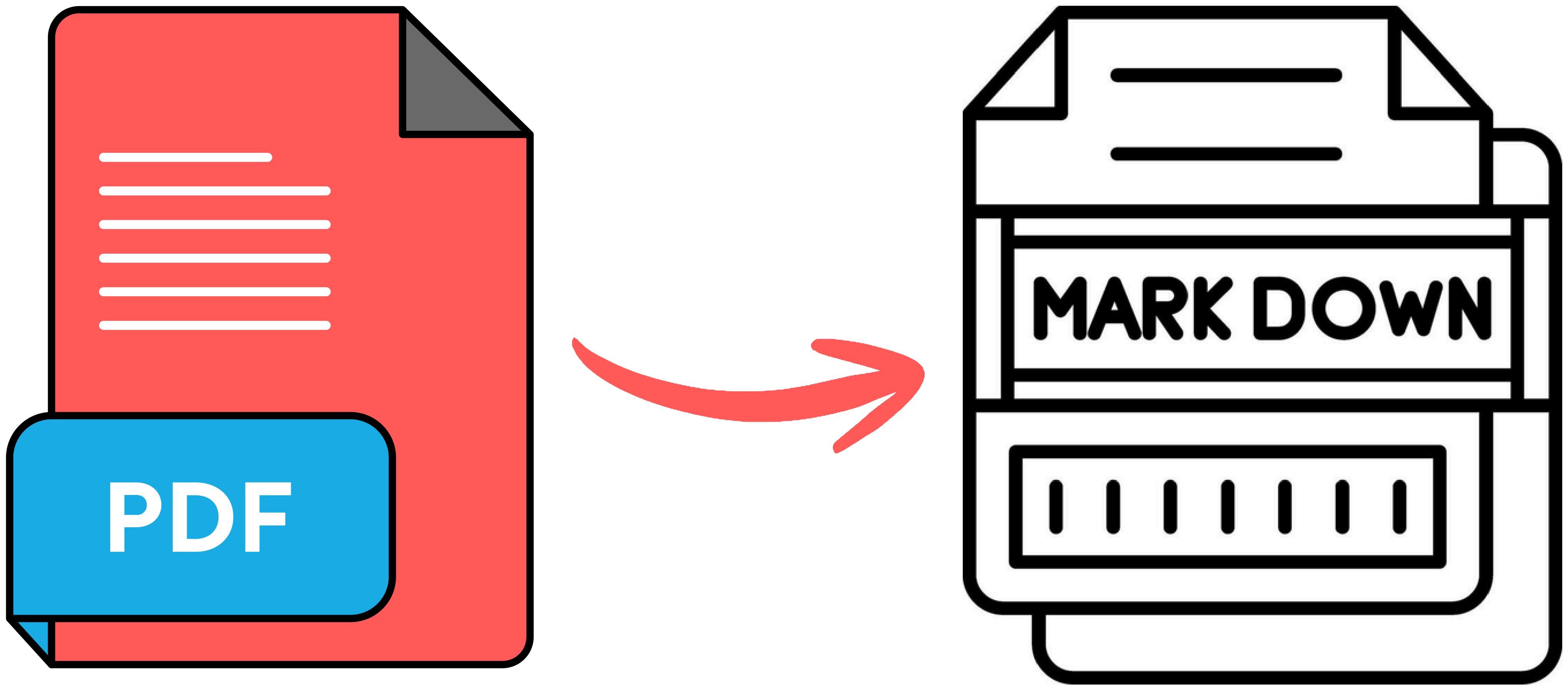


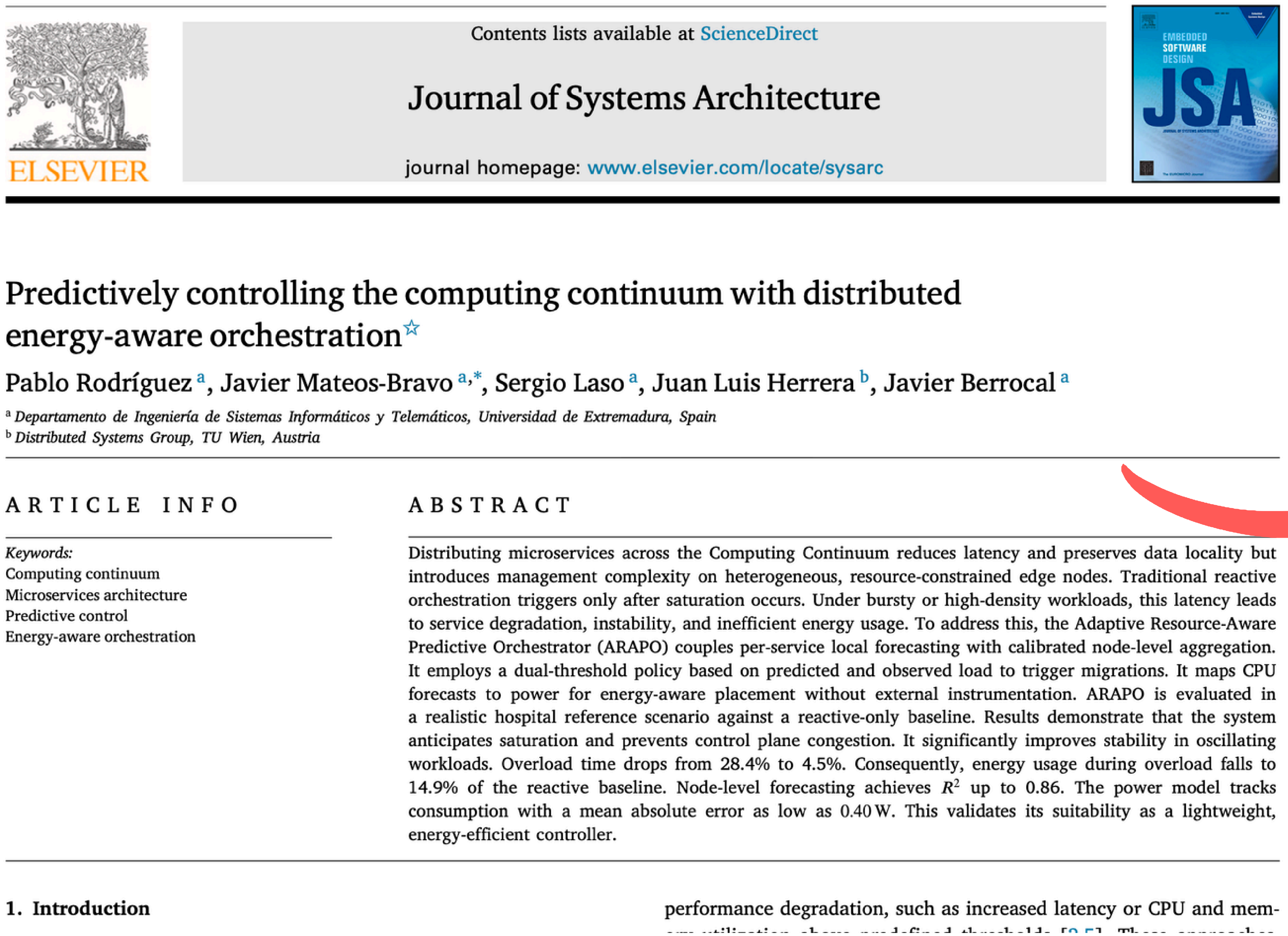

Input (texto)
Limitado por la ventana de contexto
Respuesta


Pregunta Usuario +
PDF
PDF



- Extremadamente caro
- A pocos PDFs que añadamos, no cabrán en la ventana de contexto
- Muy lento
- Demasiada información de golpe


PDF
MARK DOWN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Predictively controlling the computing continuum with distributed energy-aware orchestration☆

Pablo Rodríguez [a], Javier Mateos-Bravo [a,*], Sergio Laso [a], Juan Luis Herrera [b], Javier Berrocal [a]

[a] *Departamento de Ingeniería de Sistemas Informáticos y Telemáticos, Universidad de Extremadura, Spain*
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A R T I C L E I N F O

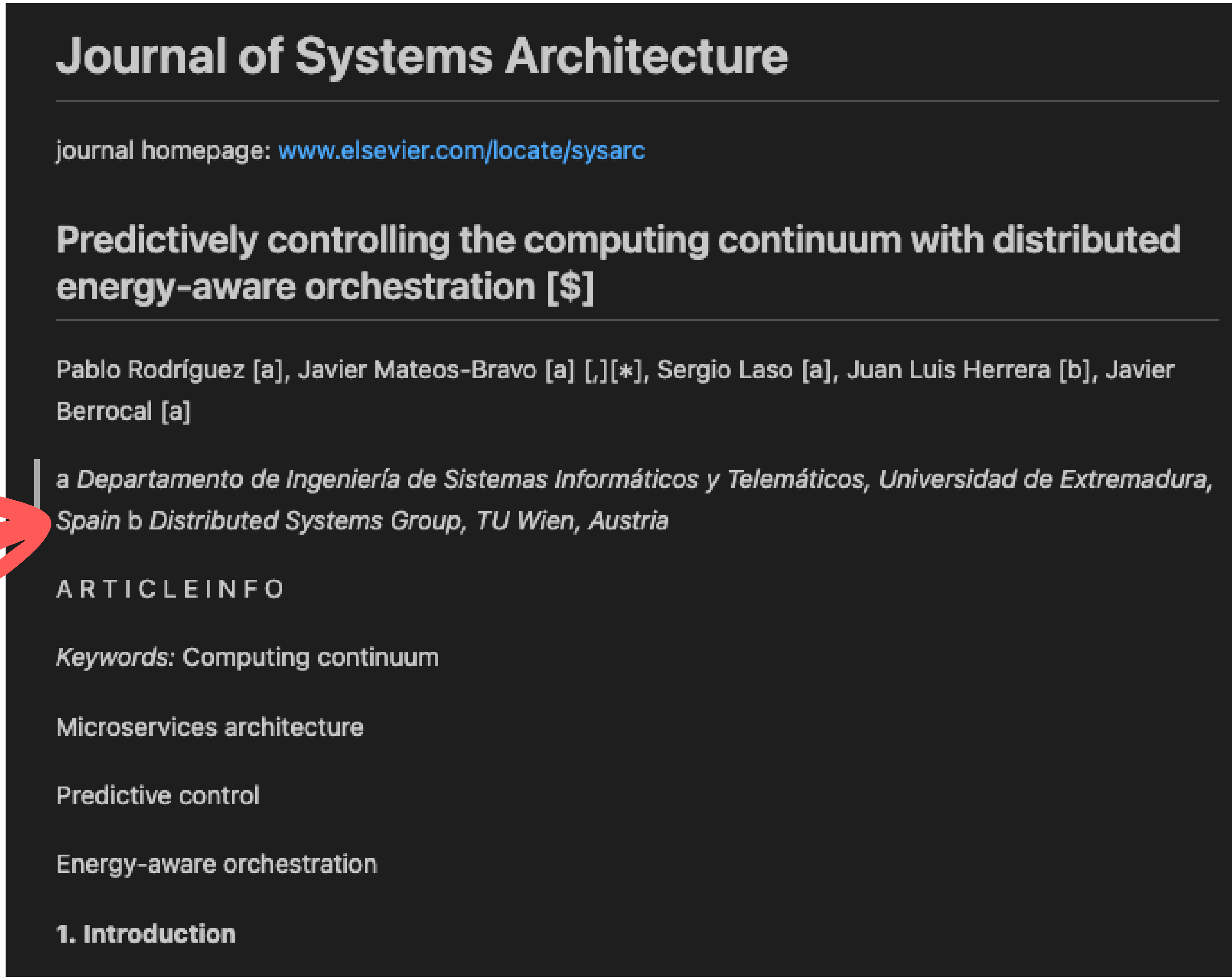
Keywords:
Computing continuum
Microservices architecture
Predictive control
Energy-aware orchestration

A B S T R A C T

Distributing microservices across the Computing Continuum reduces latency and preserves data locality but introduces management complexity on heterogeneous, resource-constrained edge nodes. Traditional reactive orchestration triggers only after saturation occurs. Under bursty or high-density workloads, this latency leads to service degradation, instability, and inefficient energy usage. To address this, the Adaptive Resource-Aware Predictive Orchestrator (ARAPO) couples per-service local forecasting with calibrated node-level aggregation. It employs a dual-threshold policy based on predicted and observed load to trigger migrations. It maps CPU forecasts to power for energy-aware placement without external instrumentation. ARAPO is evaluated in a realistic hospital reference scenario against a reactive-only baseline. Results demonstrate that the system anticipates saturation and prevents control plane congestion. It significantly improves stability in oscillating workloads. Overload time drops from 28.4% to 4.5%. Consequently, energy usage during overload falls to 14.9% of the reactive baseline. Node-level forecasting achieves R^2 up to 0.86. The power model tracks consumption with a mean absolute error as low as 0.40 W. This validates its suitability as a lightweight, energy-efficient controller.

1. Introduction

performance degradation, such as increased latency or CPU and mem-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A R T I C L E I N F O

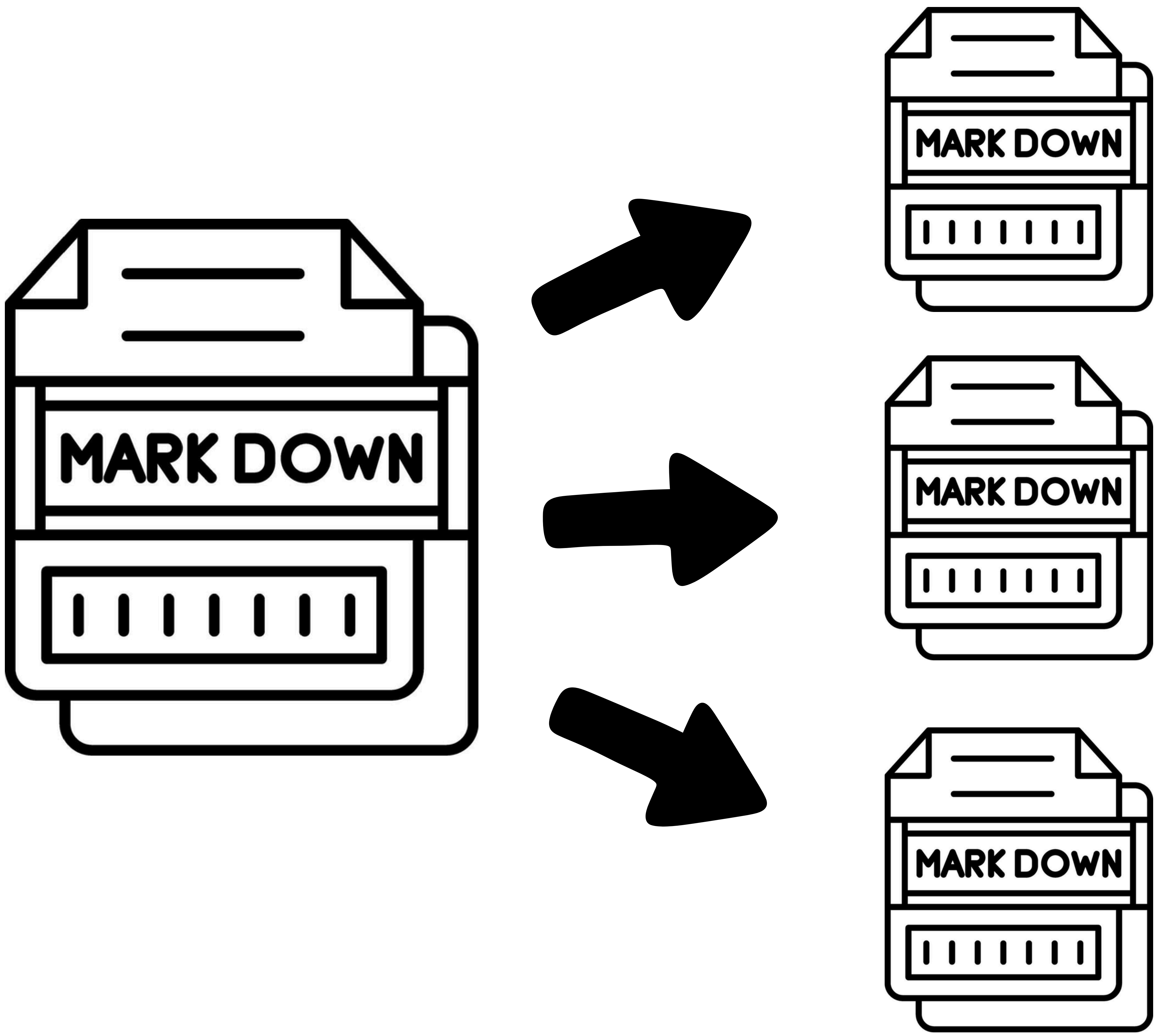
Keywords: Computing continuum

Microservices architecture

Predictive control

Energy-aware orchestration

1. Introduction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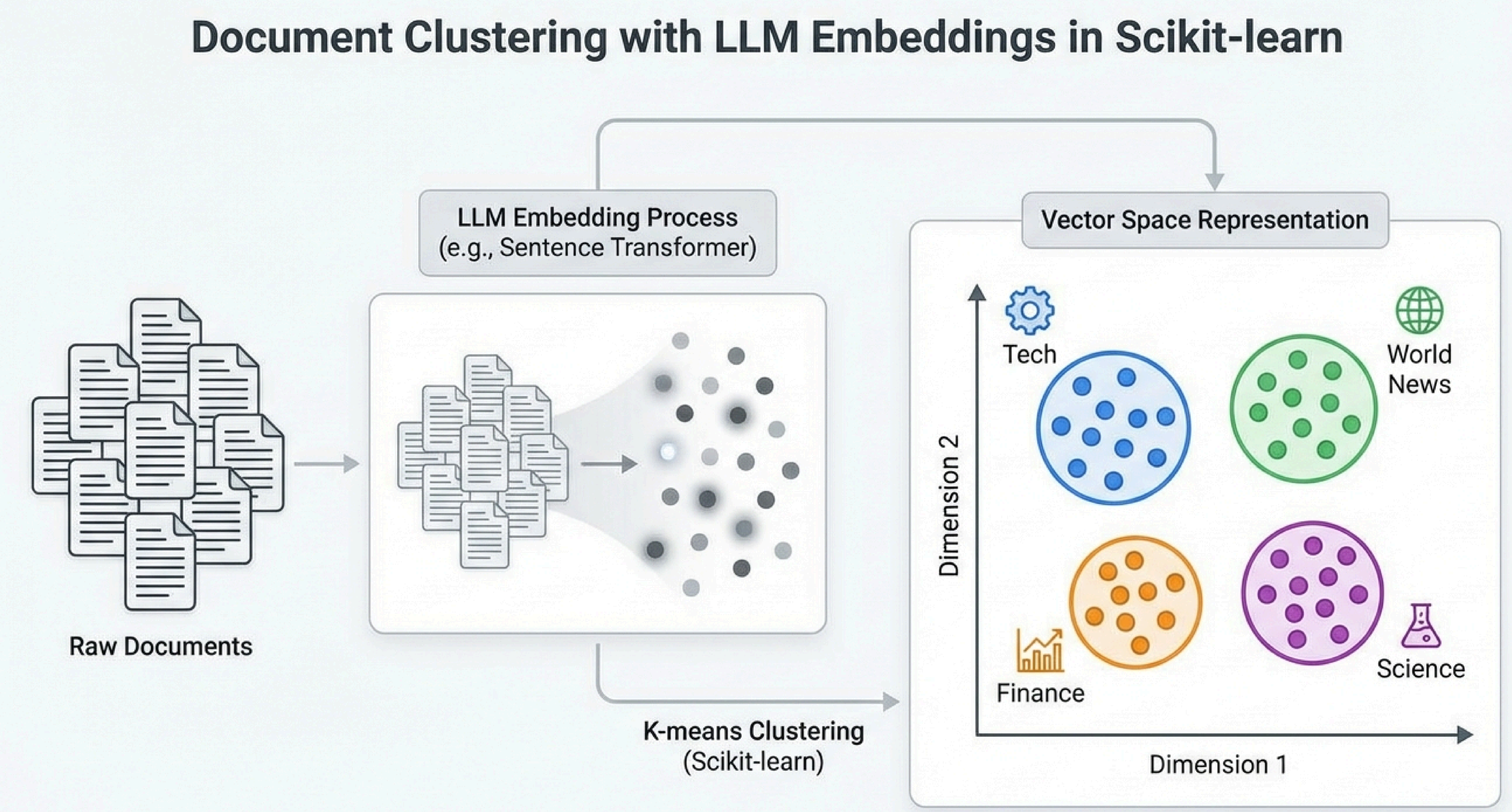
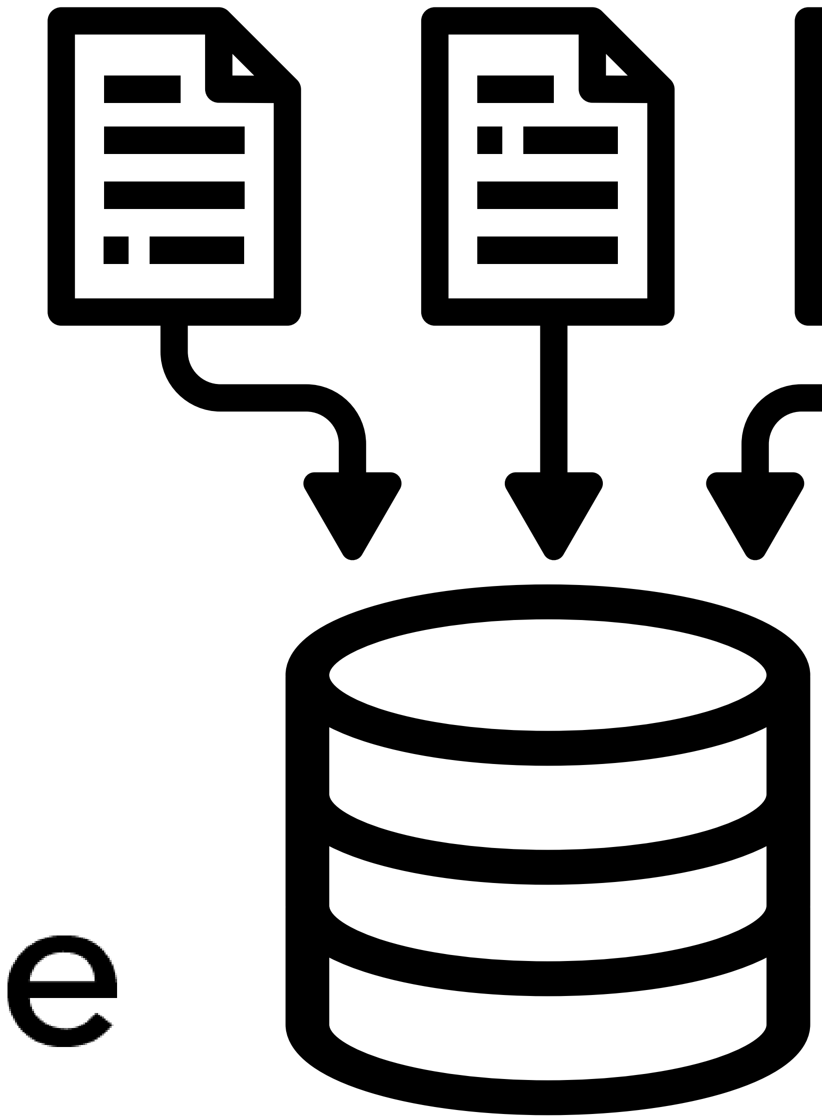

Document Clustering with LLM Embeddings in Scikit-learn
LLM Embedding Process
(e.g., Sentence Transformer)
Vector Space Representation
Raw Documents
Tech
World
News
Finance
Science
Dimension 2
Dimension 1
K-means Clustering
(Scikit-learn)

Base de Datos Vectorial

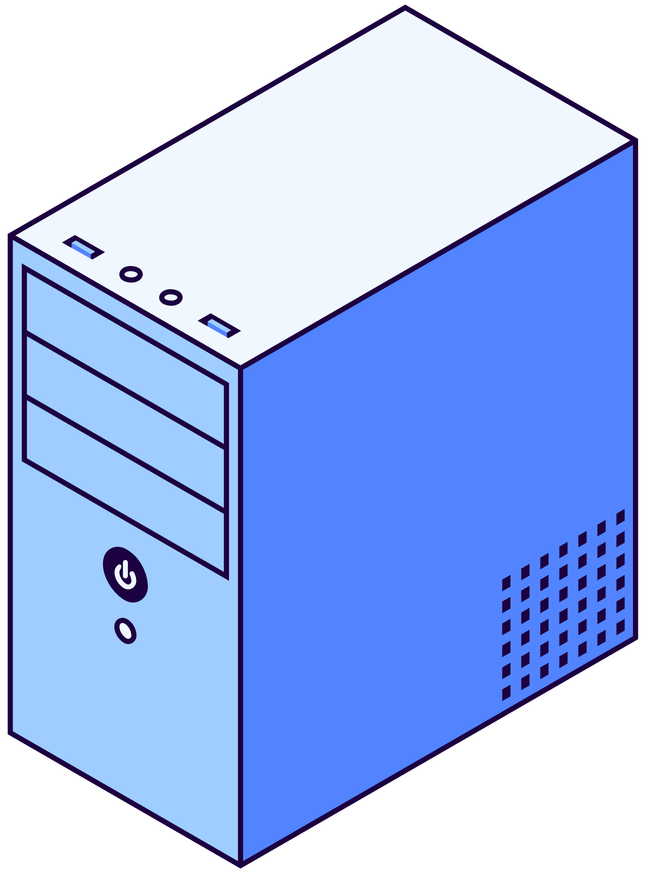
CHROMADB



milvus

Pinecone

Pgvector



MCP
(Model Context Protocol)